

✓ Datasets loaded successfully!

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📊 DATA EXPLORATION

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♦ CUSTOMER DATASET:

	Customer_ID	Tenure	MonthlyCharges	TotalCharges	Contract \
0	C00001	6	64	1540	One year
1	C00002	21	113	1753	Month-to-month
2	C00003	27	31	1455	Two year
3	C00004	53	29	7150	Month-to-month
4	C00005	16	185	1023	One year
..	...	...	...	...	...
495	C00496	50	128	6970	Month-to-month
496	C00497	2	49	5917	Two year
497	C00498	46	198	2158	Month-to-month
498	C00499	1	185	5755	Two year
499	C00500	27	181	4292	Month-to-month

	PaymentMethod	PaperlessBilling	SeniorCitizen	Churn
0	Credit Card	No	1	0
1	Electronic Check	Yes	1	0
2	Credit Card	No	1	0
3	Electronic Check	No	1	0
4	Electronic Check	No	1	0
..	...	...	...	...
495	Credit Card	Yes	1	0
496	Bank Transfer	No	1	0
497	Bank Transfer	No	0	0
498	Credit Card	Yes	1	1
499	Bank Transfer	No	1	0

[500 rows x 9 columns]

◆ SALES DATASET:

	Date	Product	Quantity	Price	Customer_ID	Region	Total_Sales
0	2024-01-01	Phone	7	37300	CUST001	East	261100
1	2024-01-02	Headphones	4	15406	CUST002	North	61624
2	2024-01-03	Phone	2	21746	CUST003	West	43492
3	2024-01-04	Headphones	1	30895	CUST004	East	30895
4	2024-01-05	Laptop	8	39835	CUST005	North	318680
..	...	...	...	...	...	...	...
95	2024-04-05	Tablet	8	20770	CUST096	North	166160
96	2024-04-06	Headphones	1	7647	CUST097	West	7647
97	2024-04-07	Tablet	5	27196	CUST098	East	135980
98	2024-04-08	Monitor	1	30717	CUST099	North	30717
99	2024-04-09	Headphones	5	23376	CUST100	South	116880

[100 rows x 7 columns]

◆ SALES DATASET SHAPE:

(100, 7)

◆ MISSING VALUES:

```
Date          0
Product       0
Quantity      0
Price         0
Customer_ID   0
Region        0
Total_Sales   0
dtype: int64
```

🔪 CLEANING DATA...

SALES DATASET COLUMNS:

```
['Date', 'Product', 'Quantity', 'Price', 'Customer_ID', 'Region', 'Total_Sales', 'Revenue']
```

CUSTOMER DATASET COLUMNS:

```
['Customer_ID', 'Tenure', 'MonthlyCharges', 'TotalCharges', 'Contract', 'PaymentMethod', 'PaperlessBilling', 'SeniorCitizen', 'Churn']
```

✅ Datasets merged successfully!

```
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👤 CUSTOMER ANALYSIS
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```
🏆 TOP 10 MOST VALUABLE CUSTOMERS:
```

```
Customer_ID
CUST016    373932
CUST007    363870
CUST083    350888
CUST073    349510
CUST020    333992
CUST084    324144
CUST070    318762
CUST005    318680
CUST065    312564
CUST028    304465
Name: Revenue, dtype: int64
```

```
🕯️ CUSTOMER LIFETIME VALUE:
```

	sum	mean	count
Customer_ID			
CUST001	261100	261100.0	1
CUST002	61624	61624.0	1
CUST003	43492	43492.0	1
CUST004	30895	30895.0	1
CUST005	318680	318680.0	1

```
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🌐 REGIONAL SALES ANALYSIS
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```

```
📍 SALES BY REGION:
```

```
Region
North    3983635
South    3737852
East     2519639
West     2123922
Name: Revenue, dtype: int64
=====
```

```
🛒 PRODUCT ANALYSIS
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```

```
🔥 BEST-SELLING PRODUCTS:
```

```
Product
Laptop      136
Tablet      127
Phone       101
Monitor      66
Headphones   48
Name: Quantity, dtype: int64
```

🔗 PRODUCTS SOLD TOGETHER

📊 SAMPLE PRODUCT COMBINATIONS:

```
Product
Headphones  [Headphones, Headphones, Headphones, Headphone...
Laptop      [Laptop, Laptop, Laptop, Laptop, Laptop, Lapto...
Monitor     [Monitor, Monitor, Monitor, Monitor, Monitor, ...
Phone       [Phone, Phone, Phone, Phone, Phone, Phone, Pho...
Tablet      [Tablet, Tablet, Tablet, Tablet, Tablet, Table...
Name: Product, dtype: object
```

📈 SEASONAL TREND ANALYSIS

📊 PIVOT TABLE ANALYSIS

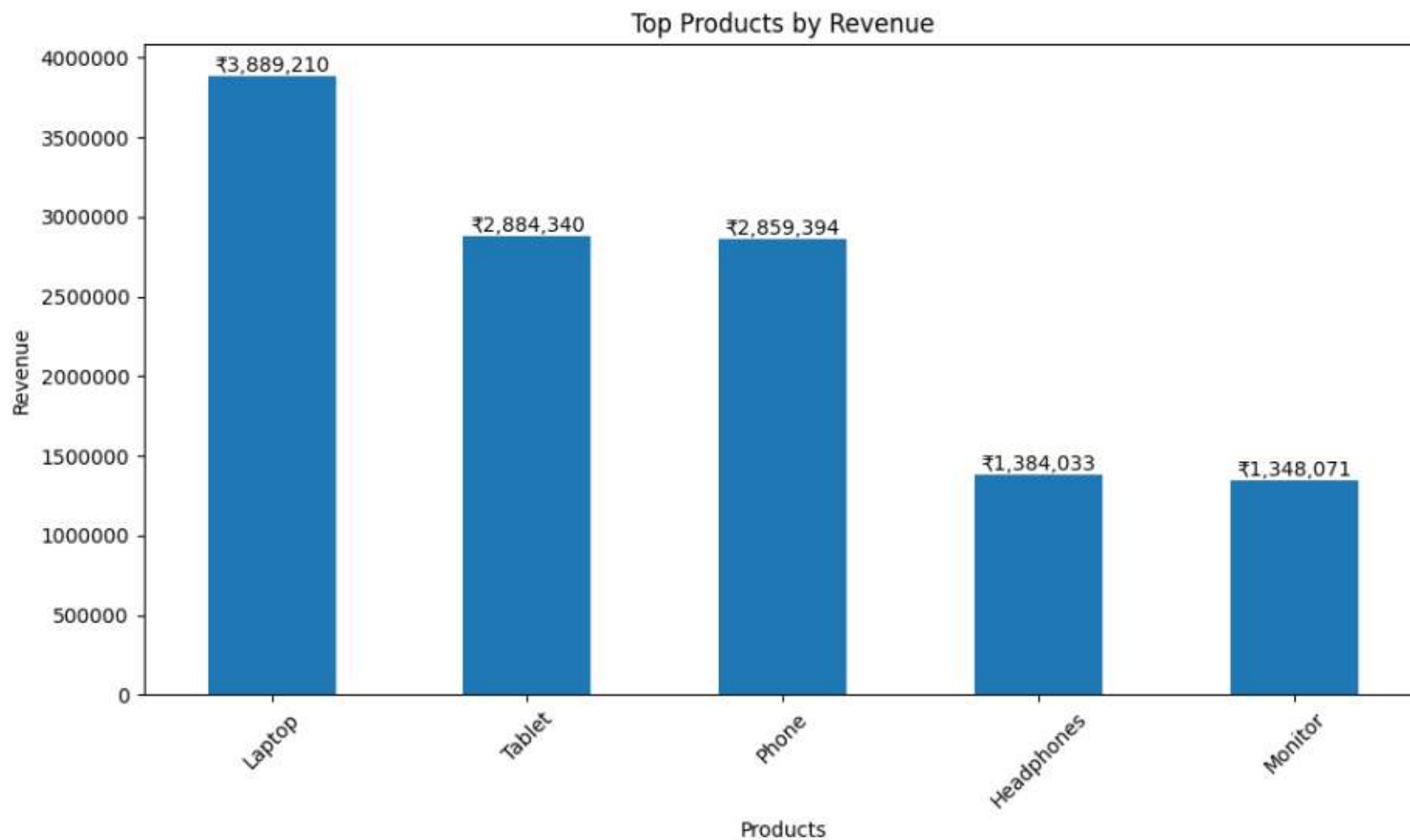
📌 REVENUE PIVOT TABLE:

Product	Headphones	Laptop	Monitor	Phone	Tablet
Region					
East	288361	221946	642870	506828	859634
North	107091	1798206	397100	489284	1191954
South	512168	1373120	39924	1471428	341212
West	476413	495938	268177	391854	491540

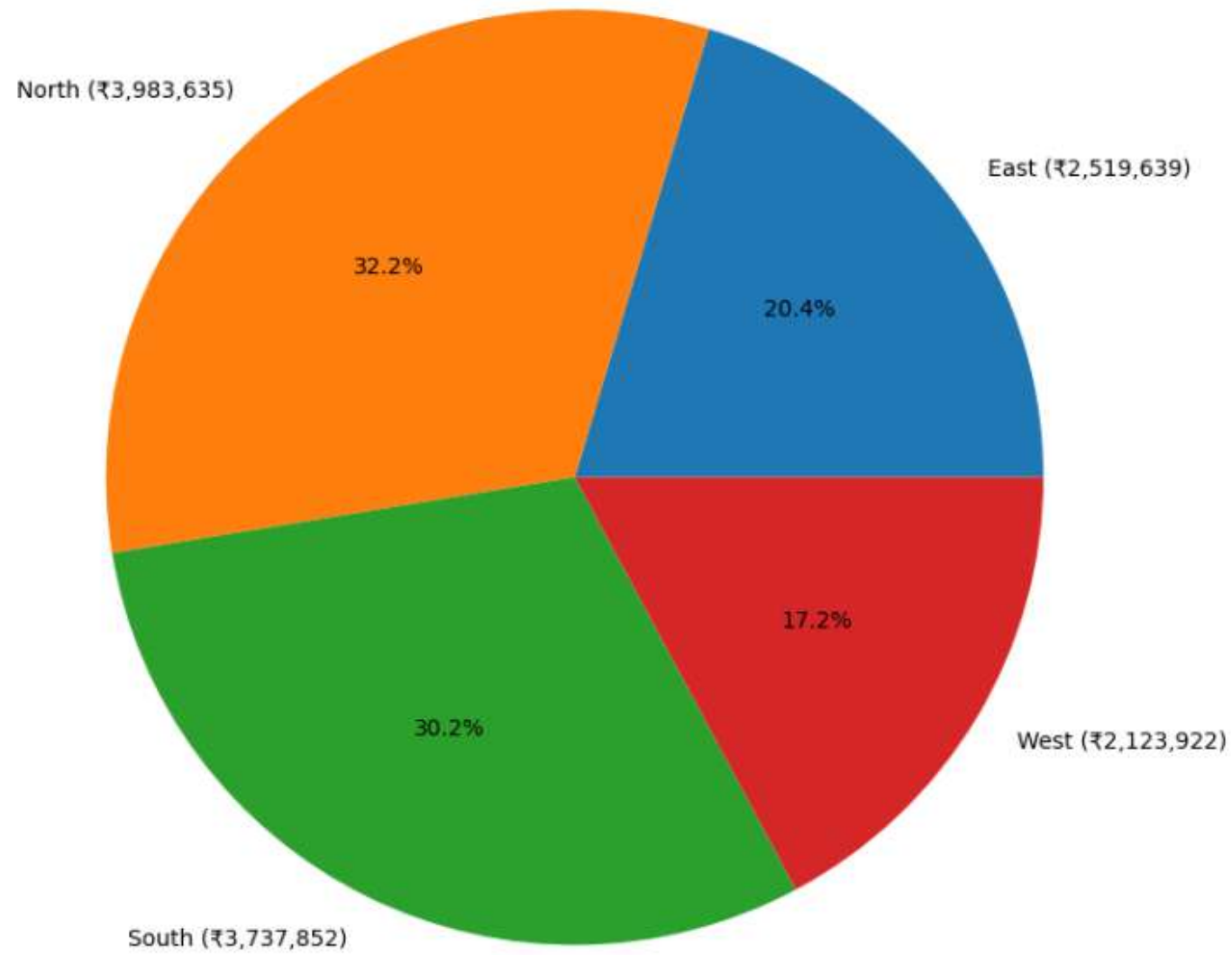
CUSTOMER RETENTION ANALYSIS

CUSTOMER RETENTION RATE: 0.00%

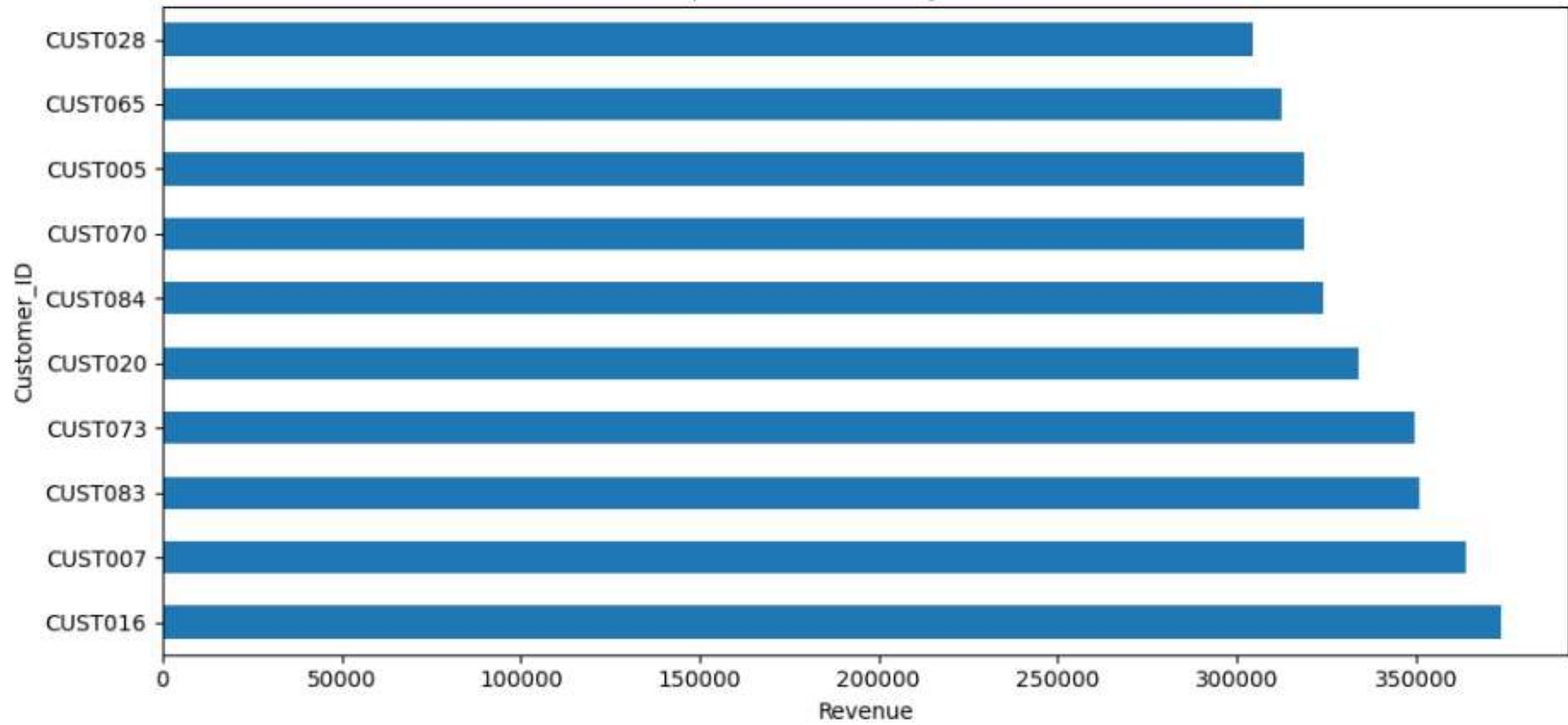
GENERATING VISUALIZATIONS...



Regional Sales Distribution



Top 10 Customers by Revenue



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🔴 BUSINESS INSIGHTS

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1. The most valuable customers contribute significantly to total revenue.
2. The highest-performing products generate the majority of sales revenue.
3. Regional analysis helps identify markets with the strongest business performance.
4. Seasonal trends reveal peak sales periods and customer buying behavior.
- # 5. Customer retention rate is 0%,
which helps evaluate customer loyalty.
6. Product combinations reveal cross-selling opportunities.
7. Pivot table analysis helps summarize large-scale business data efficiently.
8. These insights can improve:
 1. Customer retention
 2. Marketing strategy
 3. Inventory planning
 4. Product recommendations
 5. Regional targeting

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✅ CUSTOMER & SALES DATA ANALYSIS COMPLETED

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